Additional File 5

Specification of articles and moral reasons

Additional table 1: Moral reasons categories.

Moral reasons	n	References
Maximise population health	**118 (48.6%)**	
- OMPs do not maximise health of the population	63 (25.9%)	(1–63)
+ Individual needs should also be considered	6 (2.5%)	(40, 41, 64, 7, 65, 60)
- Covering OMPs means less money for common disorders	48 (19.8%)	(66, 4, 5, 61, 67, 68, 11, 69, 13, 70, 14, 71–73, 19, 62, 74–81, 29, 30, 82, 33, 83, 84, 36, 85, 86, 41, 87, 48, 50, 88, 54, 89, 57, 90–93, 60, 1, 7)
+ Low budget impact due to low prevalence	37 (15.2%)	(2, 94–96, 8, 10, 12, 97, 14, 15, 98, 72, 73, 18, 19, 21, 99–101, 33, 83, 84, 42, 45, 87, 102, 103, 88, 104, 105, 54, 106, 89, 107, 57, 92, 108)
- Taking all OMPs together, the budget impact is significant	27 (11.1%)	(109, 61, 69, 13, 73, 110, 111, 76, 77, 79, 112, 81, 113, 33, 83, 36, 41, 87, 47, 49, 114–116, 91, 117, 1, 118)
Equality and equity	**194 (79.8%)**	
- Special status of OMPs contradicts egalitarian principles	32 (13.2%)	(2, 119, 5, 120, 8, 67, 121, 69, 13, 18, 122, 76, 79, 29, 112, 101, 83, 38, 123, 43, 47, 50, 124, 125, 115, 54, 106, 55–57, 126, 1)
+ Special status of OMPs justified for reasons of equity	38 (15.6%)	(127, 5, 128, 129, 12, 97, 130, 122, 21, 76, 131, 33, 35, 85, 63, 123, 86, 41, 132, 50, 105, 54, 106, 90, 91, 133–135, 1)
+ RD patients should have the same access to treatment as others	78 (32.1%)	(136, 127, 3, 137, 138, 4, 139, 6, 109, 95, 96, 8–10, 121, 140, 69, 70, 97, 141, 142, 98, 143, 17, 144, 72, 145, 62, 110, 146, 74, 75, 147, 24, 27, 28, 112, 148, 81, 149, 30, 150–152, 34, 153, 84, 154, 155, 38, 39, 123, 156, 40, 64, 132, 87, 46, 157, 158, 47, 48, 159, 49, 160, 50, 65, 161–163, 124, 164, 56, 57, 126, 165, 58, 166)
+ All should have equal opportunities to good health	30 (12.3%)	(167, 61, 13, 97, 168, 73, 18, 122, 62, 145, 74, 147, 23, 78, 112, 33, 35, 36, 59, 37, 41, 64, 50, 162, 106, 57, 116, 126, 93, 1)
+ Prioritize OMPs to aim for equal health outcomes	6 (2.5%)	(13, 145, 74, 38, 156, 1)
+ Everybody has same right to a decent minimum standard of care	14 (5.8%)	(13, 15, 168, 18, 19, 74, 23, 30, 83, 84, 59, 41, 50, 49)
+ Prioritise those who are worse off	24 (9.9%)	(4, 7, 13, 70, 168, 122, 74, 77, 78, 36, 38, 39, 56, 169, 1, 170, 171, 12, 91, 37, 41, 42, 132, 48)
+ Prioritize OMPs because/if they treat severe rare diseases	95 (39.1%)	(172, 167, 8, 173, 174, 10, 175, 12, 98, 18, 21, 176, 76, 22, 23, 99, 177–179, 26, 112, 149, 101, 150, 83, 84, 36, 59, 86, 156, 180, 132, 87, 181, 47, 182, 48, 164, 88, 52, 104, 56, 126, 91, 92, 134, 165, 1, 127, 183, 184, 19, 62, 185, 186, 3, 138, 4, 187, 6, 120, 109, 61, 9, 11, 15, 71, 17, 73, 188, 20, 189, 75, 24, 78, 55, 148, 81, 190, 30, 57, 191, 37, 38, 40–42, 93, 43, 46, 192, 58, 49, 193)

+ Prioritize RD patients with unmet needs	29 (11.9%)	(2, 172, 68, 98, 176, 99, 24, 177, 179, 194, 195, 150, 196, 83, 36, 85, 156, 43, 45, 197, 182, 48, 198, 161, 105, 71, 56, 91, 58)
+ Prioritize OMPs when no alternative treatment is available	68 (28%)	(3, 138, 172, 120, 109, 128, 173, 174, 10, 175, 121, 11, 12, 97, 98, 17, 199, 188, 122, 111, 21, 146, 176, 23, 99, 177, 186, 27, 80, 29, 112, 81, 149, 31, 150, 191, 83–85, 38, 156, 180, 200, 132, 43, 45, 201, 197, 47, 202, 182, 48, 184, 193, 65, 187, 52, 53, 104, 189, 56, 91, 185, 92, 133, 117, 134, 165)
+ Special status because OMPs are too expensive for self-payment	22 (9.1%)	(173–175, 12, 17, 178, 28, 183, 155, 85, 132, 87, 203, 182, 102, 184, 204, 205, 92, 117, 134, 108)
+ RD patients are disadvantaged due to rarity	17 (7%)	(136, 66, 9, 206, 122, 145, 147, 77, 207, 208, 151, 63, 86, 41, 184, 1, 57)
Personal responsibility	**9 (3.7%)**	
+ RD patients are sick out of bad luck	4 (1.6%)	(13, 149, 106, 57)
Society values individual lifestyle choices	5 (2.1%)	(57, 41, 42, 182, 93)
Rule of Rescue	**62 (25.5%)**	
+ Moral intuition to rescue the identifiable no matter cost	48 (19.8%)	(209, 94, 4, 210, 172, 109, 8, 121, 13–15, 168, 73, 18–20, 147, 77, 78, 207, 25, 27, 30, 31, 191, 34, 83, 153, 211, 212, 85, 38, 39, 123, 41, 42, 7, 43, 45, 46, 159, 50, 51, 107, 52, 92, 60, 1)
- Concept of the Rule of Rescue is inappropriate	20 (8.2%)	(61, 121, 12, 168, 20, 76, 99, 77, 78, 25, 28, 80, 208, 113, 33, 43, 7, 88, 52, 1)
- The aspect of identifiability is not acceptable	21 (8.6%)	(61, 168, 74, 77, 78, 207, 80, 29, 30, 33, 83, 40, 41, 1, 50, 7, 51, 88, 92, 60, 49)
Duty	**63 (25.9)**	
+ Society has a duty to provide effective treatment if available‘	28 (11.5%)	(127, 213, 214, 139, 215, 96, 168, 145, 76, 23, 84, 35, 36, 59, 216, 40, 46, 192, 48, 160, 217, 50, 161, 105, 106, 57, 91, 93)
+ OMP reimbursement conforms to principle of social solidarity	21 (8.6%)	(137, 218, 95, 173, 206, 14, 130, 76, 78, 37, 180, 219, 41, 132, 182, 50, 105, 189, 57)
+ Receiving appropriate treatment is a human right	20 (8.2%)	(94, 10, 12, 70, 97, 130, 62, 113, 150, 33, 83, 38, 219, 41, 44, 46, 220, 221, 117, 222)
+ Doctor's duty to provide treatment in the interest of patients	8 (3.3%)	(210, 9, 62, 20, 76, 27, 46, 106)
Rarity	**77 (31.7%)**	
+ Rarity in itself is a factor that warrants special status	23 (9.5%)	(127, 61, 96, 11, 72, 188, 78, 195, 180, 41, 42, 132, 87, 223, 182, 224, 220, 125, 104, 56, 221, 91, 1)
+ Societal preference for special status for rare diseases	17 (7%)	(4, 12, 14, 18, 188, 122, 145, 101, 83, 84, 37, 219, 225, 189, 56, 1)
- Rarity in itself does not justify special status	19 (7.8%)	(9, 69, 122, 74, 99, 29, 226, 83, 154, 86, 182, 1, 198, 193, 89, 118, 126, 221)
- No societal preference for rarity alone	42 (17.3%)	(127, 227, 138, 4, 5, 172, 61, 171, 8–10, 121, 68, 11, 12, 62, 145, 20, 179, 186, 81, 80, 228, 30, 101, 191, 83, 154, 37, 85, 41, 132, 43, 47, 182, 48, 198, 49, 187, 56, 229, 1)

Additional File 5

Appropriateness of cost-effectiveness criteria	**103 (42.4%)**	
- OMPs can & should meet same cost-effec criteria as other drugs	19 (7.8%)	(4, 120, 96, 11, 73, 19, 74, 80, 29, 112, 83, 59, 41, 47, 51, 103, 104, 54, 166)
+ OMPs unlikely to meet traditional cost-effec criteria	52 (21.4%)	(127, 2, 227, 4, 172, 174, 173, 175, 9, 129, 121, 11, 97, 168, 122, 110, 21, 230, 231, 24, 78, 232, 26, 194, 195, 29, 31, 32, 113, 150, 35, 85, 123, 180, 219, 40, 41, 7, 43, 87, 182, 198, 114, 125, 89, 56, 126, 91, 233, 133, 58, 1)
+ Standard economic assessment inappropriate for OMPs	54 (22.2%)	(127, 218, 227, 4, 61, 8, 10–12, 69, 70, 98, 72, 73, 19, 122, 145, 111, 21, 146, 230, 76, 78, 177, 26, 234, 190, 101, 150, 83, 235, 37, 123, 40, 42, 236, 64, 43, 182, 48, 237, 7, 114, 88, 164, 115, 238, 104, 54, 89, 107, 185, 117, 58)
+ Reimburse if certain standards are met	25 (10.3%)	(120, 61, 67, 121, 13, 97, 98, 73, 111, 21, 146, 176, 78, 191, 83, 219, 40, 64, 182, 198, 51, 105, 89, 56, 57)
Treatment benefit	**76 (31.3%)**	
+ Benefits of OMPs justify special status	48 (19.8%)	(94, 138, 172, 167, 96, 9–13, 97, 98, 73, 188, 122, 62, 20, 176, 76, 99, 78, 177, 100, 239, 186, 81, 190, 30, 101, 150, 83, 84, 235, 85, 123, 156, 40, 132, 47, 182, 48, 224, 105, 89, 189, 56, 57, 240)
+ OMPs can lower social/economic burden of society	25 (10.3%)	(139, 167, 70, 62, 110, 21, 146, 26, 100, 239, 190, 83, 84, 63, 236, 87, 48, 184, 65, 224, 189, 91, 185, 92, 240)
OMP should have proven effectiveness & safety	26 (10.7%)	(2, 215, 97, 206, 15, 98, 111, 146, 176, 232, 29, 112, 83, 35, 85, 40, 181, 47, 182, 48, 7, 163, 105, 57, 165, 166)
R&D investments	**49 (20.2%)**	
+ Reimbursing OMPs incentivises more R&D in the future	38 (15.6%)	(96, 171, 9, 140, 97, 72, 18, 21, 176, 179, 194, 239, 27, 28, 112, 196, 83, 84, 216, 43, 87, 223, 184, 193, 217, 65, 224, 54, 106, 89, 189, 107, 56, 241, 57, 91, 117, 165)
+ Duty to reimburse OMPs because so much investment into R&D	19 (7.8%)	(213, 218, 96, 11, 12, 97, 242, 243, 112, 81, 30, 208, 83, 84, 132, 223, 224, 57, 125)
Total no of documents	243 (100%)	

Notes: n = number of documents this reason has been mentioned. + = Reason in favor of special status of OMPs. – = Reason against special status of OMPs.

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